

TFPT — Appendix H: the horizon unit system

One seam constant $c_3 = \frac{1}{8\pi}$ as the universal horizon thermal code
(Hawking, de Sitter, Unruh, Page, scrambling — *standard physics in seam units, not new gravity*)

June 11, 2026

What this document is about

A change of bookkeeping, not new gravitational physics: if gravity is the geometry-channel readout of the seam, then *all* horizons read the same boundary constant $c_3 = \frac{1}{8\pi}$. This note collects the readouts — Hawking, de Sitter and Unruh temperature, black-hole thermodynamics, the Page time, scrambling, the Nariai bound, $v_{\text{GW}} = c$ and cosmic birefringence — in seam units, with two genuine compiler fingerprints ($1920 = |W(D_5)|$, $|\mu_4| = 4$).

The TFPT document set — what is treated where

Plain language: TFPT is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard Model, the constants and the scale grammar. The development is **five short documents plus this appendix**, best read in order:

1. **introduction** — reading guide, compiler closure, paper-by-paper comparison, predictions, the dependency DAG and proof ledger.
2. **tfpt_1_architecture_e8** — the two axioms, the derivation map, the EM fixed point α^{-1} , the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction.
3. **tfpt_2_standard_model** — the SM in one φ_0 -ladder formula, flavor from parabolic transport, the worked closures, and gravity/QG as the seam response.
4. **tfpt_3_e8_audit_bootstrap** — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
5. **tfpt_4_frontier** — honest status of η_B , m_p/m_e , Koide, dark matter, full QG.
- H. **tfpt_horizon_readouts** — **Appendix H**: one seam constant as the universal horizon thermal code (a unit-system reframe, *not* a peer paper, *not* new physics).

You are reading Appendix H — a notation/unit reframing; all new-physics content lives in documents 1–4.

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Scope and honest typing

If gravity is the geometry-channel readout of the seam (companion note “Seam response grammar”), then horizons are not separate miracles — they all read the *same* boundary normaliser $c_3 = 1/(8\pi)$. This note collects the readouts. Markers: **[P]** standard horizon physics rewritten in TFPT/seam units (no new claim), **[I]** an exact compiler connection (audit-level), **[N]** a numerical readout (reproduced), **[A]** a search target, not a result. Nothing here is sold as new gravitational physics; it is a *change of bookkeeping* that exposes shared structure.

Unit conventions used throughout

Three unit systems appear; the seam factor $c_3 = 1/(8\pi)$ is the same number in all of them, only the carried dimensionful constants change:

system	convention	Hawking temperature
Planck	$G=\hbar=c=k_B=1$	$T_H = c_3/M = 1/(8\pi M)$
reduced Planck	$\bar{M}_{\text{Pl}} = M_{\text{Pl}}/\sqrt{8\pi},$ $\hbar=c=k_B=1$	$T_H = c_3 \bar{M}_{\text{Pl}}^2/M$
SI	restore $G, \hbar, c, k_B$	$T_H = \frac{\hbar c^3}{8\pi G k_B M} = c_3 \frac{\hbar c^3}{G k_B M}$

All boxed formulas below are in Planck units unless stated; the seam constant is dimensionless.

Standard references for the formulas reframed here: Hawking temperature and radiation (Hawking 1975); Bekenstein–Hawking entropy (Bekenstein 1973, Hawking 1975); Page time (Page 1993); fast scrambling $t_{\text{scr}} \sim \beta \log S$ (Sekino–Susskind 2008, Maldacena–Shenker–Stanford 2016); Nariai bound (Nariai 1950); de Sitter thermodynamics (Gibbons–Hawking 1977). *Nothing below is a new derivation of these* — only a rewriting in the seam unit c_3 .

1 The seam constant is the universal horizon temperature factor

Every Killing horizon has $T = \hbar\kappa/(2\pi c k_B)$. Since

$$\boxed{\frac{1}{2\pi} = 4c_3, \quad \frac{1}{8\pi} = c_3} \quad \text{[I]},$$

the universal factor *is* the seam constant: $T_{\text{hor}} = 4c_3 \hbar\kappa/(c k_B)$. Black holes, de Sitter and Unruh therefore share *one* thermal grammar. **[P]** [\[verification/v8_horizon.py\]](#)

2 Schwarzschild thermodynamics in four c_3 -lines

For a non-rotating, uncharged hole in Planck units ($G=\hbar=c=k_B=1$):

$$\boxed{T_H = \frac{c_3}{M}, \quad S_{\text{BH}} = \frac{M^2}{2c_3}, \quad P_H = \frac{c_3}{1920 M^2}, \quad \tau_{\text{evap}} = \frac{640}{c_3} M^3} \quad \text{[P]}.$$

Two of the constants are compiler numbers, not coincidences:

- the Hawking temperature factor is $c_3 = 1/(8\pi)$ — the same P1 boundary normaliser; [I]
- the radiated-power denominator is $1920 = |W(D_5)|$, the Weyl-group order of the D_5 carrier. [I]

Scope of P_H : this is the idealised massless single-channel blackbody normalisation. Greybody factors and the radiated species content shift the phenomenological evaporation coefficient; the c_3 -form is the convention statement, not a general phenomenological prediction.

object	T_H	lifetime / entropy
solar-mass hole	$6.17 \times 10^{-8} \text{ K}$	$\tau_{\text{evap}} \approx 2.1 \times 10^{67} \text{ yr};$ $S/k_B \approx 1.05 \times 10^{77}$
PBH evaporating now ($1.7 \times 10^{11} \text{ kg}$)	$k_B T \approx 61 \text{ MeV}$	relevant for γ/ν PBH signatures

3 Page time and scrambling

Since $S_{BH} \propto M^2$ and $\tau \propto M^3$, the Page point ($M \rightarrow M/\sqrt{2}$) is

$$t_{\text{Page}} = \left(1 - \frac{1}{2\sqrt{2}}\right) \tau_{\text{evap}} \approx 0.6464 \tau_{\text{evap}} \quad [\text{P}].$$

In TFPT the Page curve is *boundary recoverability*: the physical sector is OS-reconstructed and gapped, and the recovered mutual information decays at the transfer rate, $I(A:C|B)_n \sim \lambda_2^n = (2/3)^{6n}$. **This is the same operator that builds the Standard Model**: the boundary transport spectrum is $\{1, (2/3)^6, (1/3)^6\}$ and its sub-leading eigenvalue $\lambda_2 = (2/3)^6$ is exactly the flavor-gap eigenvalue of Paper 2 ($\Delta_{\text{gap}} = -\log(2/3)^6 = 6 \log \frac{3}{2}$). One boundary transport thus governs *both* the SM flavor hierarchy and the horizon information recovery. [I] [verification/v54_seam_horizon_keystones.py] The scrambling prefactor carries a μ_4 signature: with $\beta = 1/T_H = 8\pi M$, $\beta/2\pi = 4M$, so

$$t_{\text{scr}} \sim |\mu_4| M \log S \quad (|\mu_4| = 4) \quad [\text{I}]\text{-audit.}$$

4 De Sitter and the Nariai bound from the Λ closure

With the action-grammar Hubble/ Λ readouts ($H_0/\bar{M}_{\text{Pl}} = e^{-\alpha_\star^{-1}}/(2\pi\sqrt{\Omega_\Lambda})$, $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 \sim e^{-2\alpha_\star^{-1}}$), the asymptotic de Sitter horizon gives

$$\frac{T_{dS}}{\bar{M}_{\text{Pl}}} = 16c_3^2 e^{-\alpha_\star^{-1}}, \quad S_{dS} = \frac{e^{2\alpha_\star^{-1}}}{128c_3^4} = 32\pi^4 e^{2\alpha_\star^{-1}} \approx 3.32 \times 10^{122} \quad [\text{P}].$$

This is the area-law horizon capacity of the universe from $(\alpha_\star^{-1}, c_3, H_\Lambda)$ alone. The maximal black hole in de Sitter (Nariai) is

$$M_N \approx 2.16 \times 10^{22} M_\odot \approx 4.3 \times 10^{52} \text{ kg}, \quad S_N = \frac{1}{3} S_{dS} \quad [\text{P}],$$

the absolute gravitational-capacity limit, not an observed structure.

5 The maximal black hole is the anchor: SdS in seam units

The Nariai limit is not just a capacity number — written in compiler units, the entire Schwarzschild–de Sitter (SdS) family lands on atoms that are already load-bearing elsewhere in TFPT, with no adjustable parameter on either side. [I] [verification/v101_horizon_anchor.py]

Nariai = the anchor; the Koide 2/3 is the entropy bound [I]
 ([verification/v101_horizon_anchor.py])

With $f(r) = 1 - 2M/r - \Lambda r^2/3$, the horizon cubic at the maximal mass $M_N = 1/(3\sqrt{\Lambda}) = 1/(N_{\text{fam}}\sqrt{\Lambda})$ is, in units $1/\sqrt{\Lambda}$,

$$t^3 - 3t + 2 = (t - 1)^2(t + 2) \quad \text{roots } (1, 1, -2),$$

with coefficients $(-3, 2) = (-N_{\text{fam}}, |\mathbb{Z}_2|)$ — and the root pattern is exactly the **traceless projection of the anchor**: $a - \frac{p_1(a)}{3}\mathbf{1} = -\frac{1}{3}(1, 1, -2)$ for $a = (1, 1, 2)$ (compare $\chi_a = (t - 1)^2(t - 2)$, Paper 1). Five exact companions:

- **Entropy bound = the Koide branch value.** Each Nariai horizon carries $S_{dS}/3 = S_{dS}/N_{\text{fam}}$ and the total is $S_{\text{Nariai}}/S_{dS} = 2/3 = |\mathbb{Z}_2|/N_{\text{fam}}$ (deficit exactly $S_{dS}/3$). The interpolation in $x = r_b/r_c$ is the closed form $S_{\text{tot}}/S_{dS} = (x^2 + 1)/(x^2 + x + 1)$ — denominator = $\Phi_3(x)$, the N_{fam} cyclotomic — monotone from 1 to 2/3 with its minimum exactly at the merge (Fig. 2a).
- **Three-sheet conservation.** The cubic is traceless with fixed e_2 , so $\pi \sum_i r_i^2 = 6\pi/\Lambda = |\mathbb{Z}_2| S_{dS}$ for *every* M : the black hole redistributes entropy between the two physical sheets and the virtual third root; the three-sheet total never moves.
- **Koide-form quotient of the roots.** $Q_{\text{geom}} = \sum r_i^2 / (\sum |r_i|)^2$ runs monotonically over $[\frac{3}{8}, \frac{1}{2}]$: pure dS gives $\frac{1}{2} = |\mathbb{Z}_2|/|\mu_4| = \delta$ (the half-step), Nariai gives $\frac{3}{8} = p_2(a)/e_1(a)^2$ — exactly the two nonzero $SU(4)_1$ conformal weights (Fig. 2b).
- **The mass line is a double cover.** $\text{disc}_r \propto (1 - 3m)(1 + 3m)$ in $m = M\sqrt{\Lambda}$: branch points $m = \pm 1/N_{\text{fam}}$, separation $2/3 = |\mathbb{Z}_2|/N_{\text{fam}}$, deck involution = the black-hole $\leftrightarrow$ cosmological horizon swap — the gravitational twin of the flavor sheet $\mathbb{Z}_2$ and of the split N_{fam} -linear cover $(3x+2)(3x+5)$ of Paper 2 (Fig. 1).
- **Temperature lemma / orientation.** $|\kappa_b/\kappa_c| = (2x+1)/(x(x+2)) > 1$ for $x < 1$ (equality iff $x = 1$): the black-hole sheet is *always* hotter, so evaporation flows *away* from the merge — the anchor configuration is the **repeller** and the democratic endpoint the attractor, the same orientation as the flavor flow (Koide attractor / carrier repeller, Paper 4). *Audit reading.*

Completing the seam-unit table: $dM = c_3 \kappa dA$ (first law), $M = 2c_3 \kappa A$ (Smarr), $S \leq ER/(4c_3)$ (Bekenstein), $P_H = c_3/(1920 M^2)$, lifetime $\tau = 5120\pi M^3 = 128 g_{\text{car}} M^3/c_3$ ($128 = 2^7$ the dS prefactor, 7 the scalaron exponent; photon/geometric-optics conventions, typed external), Kerr $A_{\text{ext}} = M^2/c_3$ with $A_{\text{ext}}/A_{\text{Schw}} = 1/2 = 1/|\mathbb{Z}_2|$ and $S_{\text{ext}} = M^2/(4c_3)$, and the Bekenstein–Mukhanov–Hod area quantum $4 \ln 3 = \ln 81 = \ln(N_{\text{fam}}^4) = \ln(\text{disc of the flavor cover})$ ([P] as the QNM reading). *Honest scope*: the GR identities are textbook facts; the [I] content is six independent landings on already-load-bearing atoms with zero free parameters. The “carrier in the bulk” reading (two glue chiralities = two horizon sheets, ramification = Nariai) stays [P]. No new data test (M_N today $\sim 10^{22} M_\odot$): a structure surface, not a measurement surface.

6 Turnaround radius, gravitational waves, light

Bound structures. In a Λ -universe the maximal bound radius is $r_{\text{ta}}(M) = (GM[2\pi e^{\alpha_*^{-1}}/\bar{M}_{\text{Pl}}]^2)^{1/3}$ — halo, cluster, supercluster and void scales follow from the Λ closure with no new parameter. [P]

Gravitational waves. The low-curvature branch is $R + R^2$; the tensor graviton runs on the Lorentz cone and the only extra mode (the scalaron, $M_{\text{scal}} \approx 3.06 \times 10^{13} \text{ GeV}$) is frozen for

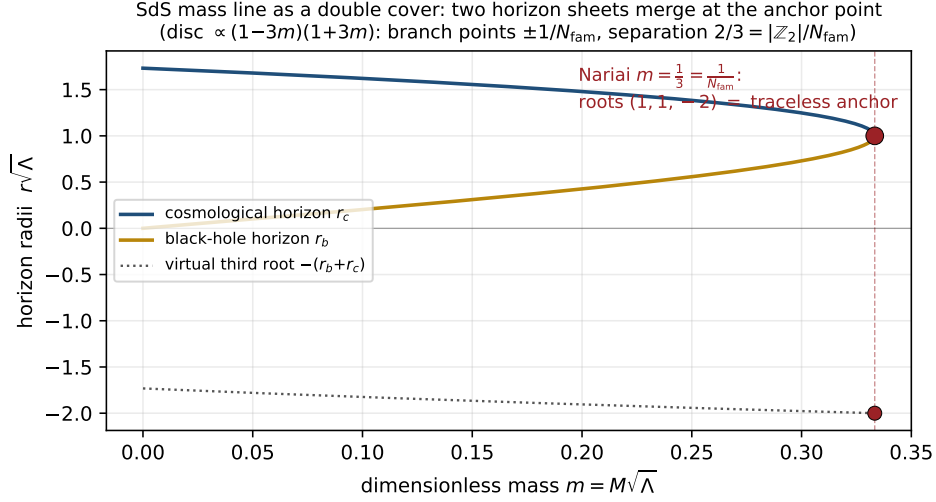


Figure 1: The SdS mass line as a double cover: black-hole and cosmological horizon sheets merge at the Nariai point $m = 1/N_{\text{fam}}$, where the three roots form the traceless anchor pattern $(1, 1, -2)$.
[\[verification/v101_horizon_anchor.py\]](#)

astrophysical frequencies. Hence

$$\boxed{v_{\text{GW}} = c} \quad (\text{no measurable dispersion}) \quad [\text{P}],$$

a clean falsifier: a genuine $v_{\text{GW}} \neq c$ would break this gravity closure.

Light. TFPT does not predict the SI number c (definitional); it predicts the *universality of the causal cone* — RP $\rightarrow$ OS reconstruction $\rightarrow$ one Lorentzian cone shared by EM, gravity and massless modes ($v_\gamma = v_g = c$, no native photon dispersion). What it *does* compute is not a speed shift but a polarisation rotation, cosmic birefringence

$$\boxed{\beta_{\text{rad}} = \frac{u}{4\pi} \approx 0.2424^\circ} \quad [\text{N}],$$

from the same seed $u = \phi_0^{\text{ret}}$ that fixes the Cabibbo angle, Ω_b and θ_{13} .

7 Search targets (not claims)

Audit-level search ansätze [A]

- **Black-hole echoes:** any near-horizon echo amplitude ratio would be a natural transfer-rate target, $\mathcal{A}_{n+1}/\mathcal{A}_n \lesssim (2/3)^6 \approx 0.0878$. A search ansatz, no prediction.
- **Quantum recovery:** the Page-curve recovery kernel $I \sim (2/3)^{6n}$ is a falsifiable shape if a boundary-recovery rate is ever measured.

Net

All horizon formulas share the one seam normalisation $1/(2\pi) = 4c_3$. Black-hole, de Sitter and Unruh temperatures, the Page time, scrambling, the Nariai bound, turnaround radii, $v_{\text{GW}} = c$ and birefringence are *readouts of one seam constant* — standard physics rewritten in TFPT units, with two genuine compiler fingerprints ($1920 = |W(D_5)|$ in Hawking power, $|\mu_4| = 4$ in scrambling). No new gravitational physics is claimed; the value of the reframe is that horizons stop being separate modules.

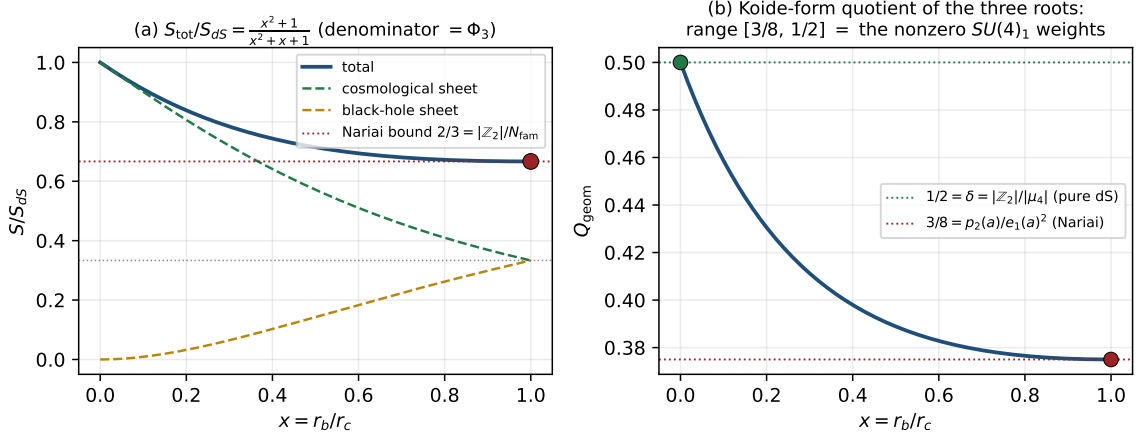


Figure 2: (a) Total SdS entropy interpolates from S_{ds} to the Nariai bound $\frac{2}{3}S_{ds}$ via $(x^2+1)/\Phi_3(x)$. (b) The Koide-form quotient of the three roots runs exactly over the nonzero $SU(4)_1$ weights $[\frac{3}{8}, \frac{1}{2}]$.
[\[verification/v101_horizon_anchor.py\]](#)

Appendix: computational verification

The seam-unit identities of this note (the thermal grammar $\frac{1}{2\pi}=4c_3$, the power denominator $1920=|W(D_5)|$, the de Sitter entropy form, the Page fraction, the scrambling $|\mu_4|=4$, and $\beta_{rad}=\varphi_0^{ret}/4\pi=0.2424^\circ$) are re-derived and machine-checked in `verification/v8_horizon.py`. Run `python verification/v8_horizon.py` (needs `mpmath`); it imports the same $c_3=1/(8\pi)$ primitive as the rest of the suite. The search-target ansätze remain [\[A\]](#) and are deliberately not part of the pass/fail checks.